

RidePulse

A Real-Time Bus Tracking
System

Project Presentation

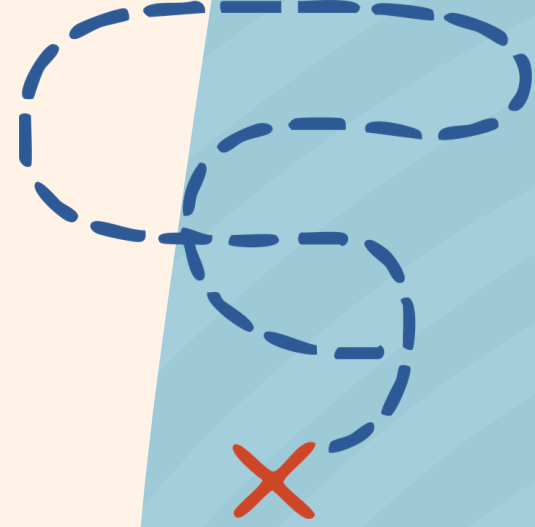
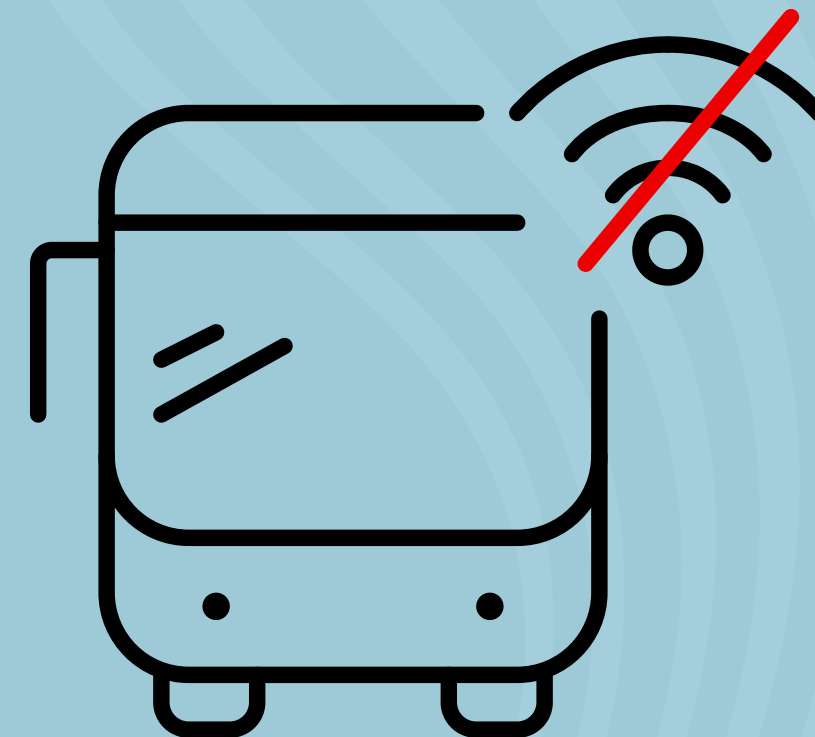
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IS-241-844-0014



THE PROBLEM



Passengers often face uncertainty while waiting for city buses because they have no access to real-time location information. This results in longer waiting times and inconvenience during daily travel.

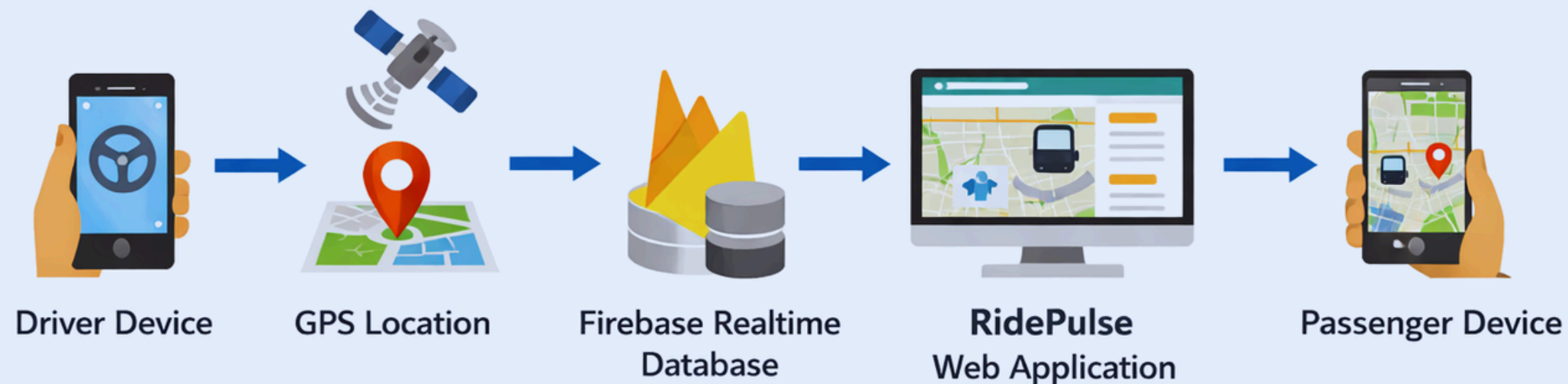


OBJECTIVES

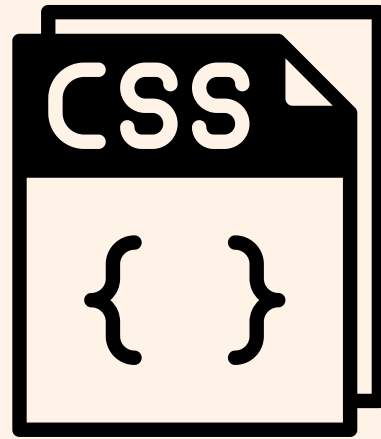
- Track buses in real time
- Reduce passenger waiting time
- Provide location updates through a web application
- Demonstrate GPS and cloud database integration



PROPOSED SOLUTION



RidePulse is a web-based bus tracking system that allows drivers to share their live location through GPS-enabled devices. The location data is synchronised through Firebase Realtime Database and displayed to passengers using an interactive map.



TECHNOLOGIES AND TOOLS

The RidePulse system was developed using modern web technologies and cloud services.

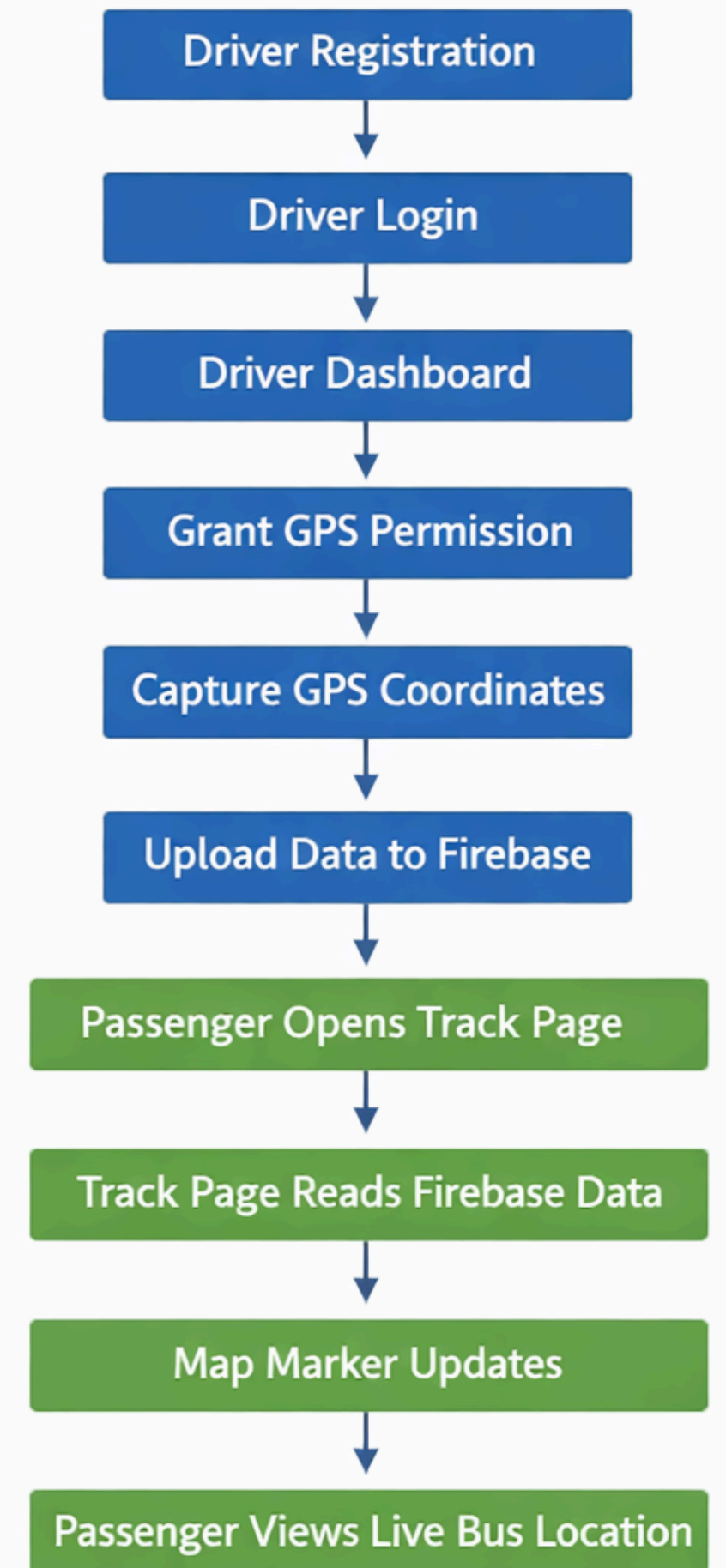
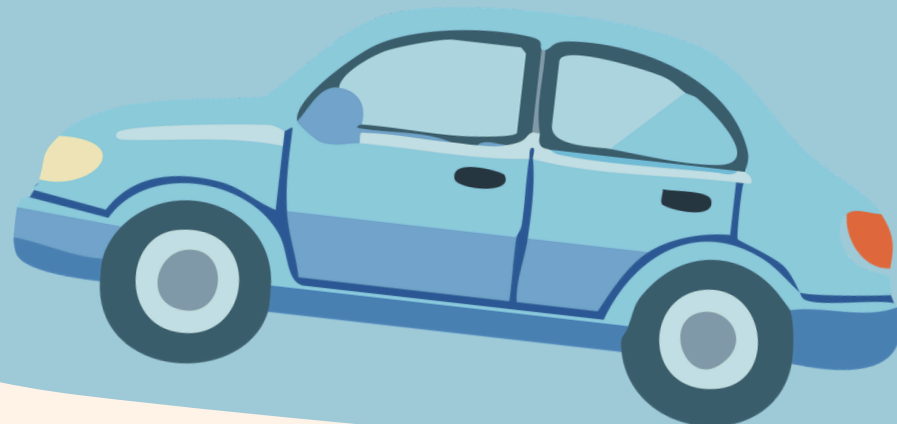
Frontend: HTML, CSS, JavaScript

Backend: Firebase Realtime Database

Additional: Geolocation API, Leaflet Maps, Vercel

SYSTEM ARCHITECTURE

The RidePulse architecture follows a client-cloud-client model. The driver device collects GPS coordinates and stores them in Firebase. Passenger devices retrieve these coordinates and display them on an interactive map interface.



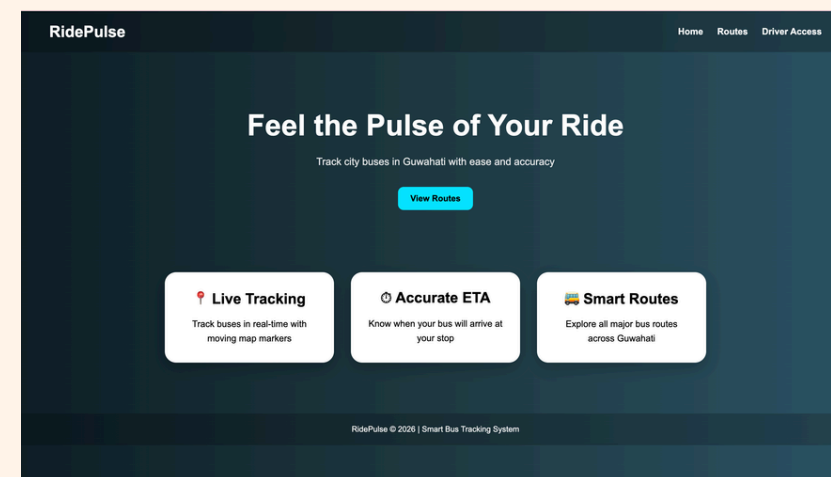


SYSTEM IMPLEMENTATION

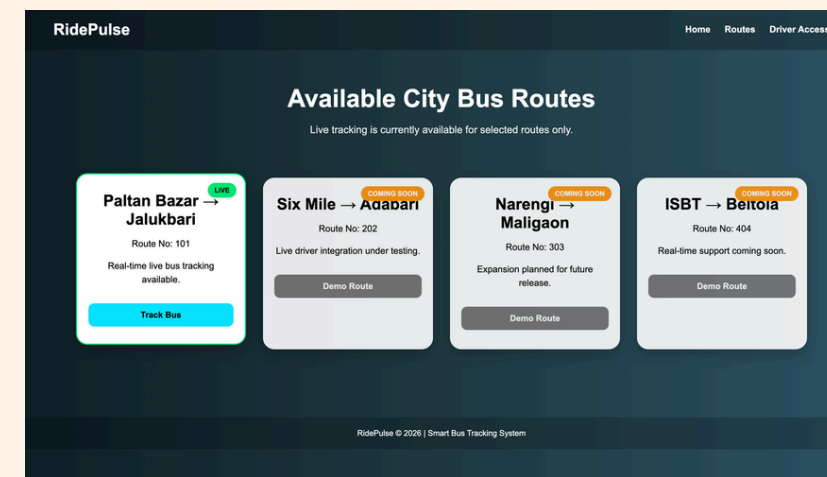
The system consists of several modules developed to support real-time bus tracking.



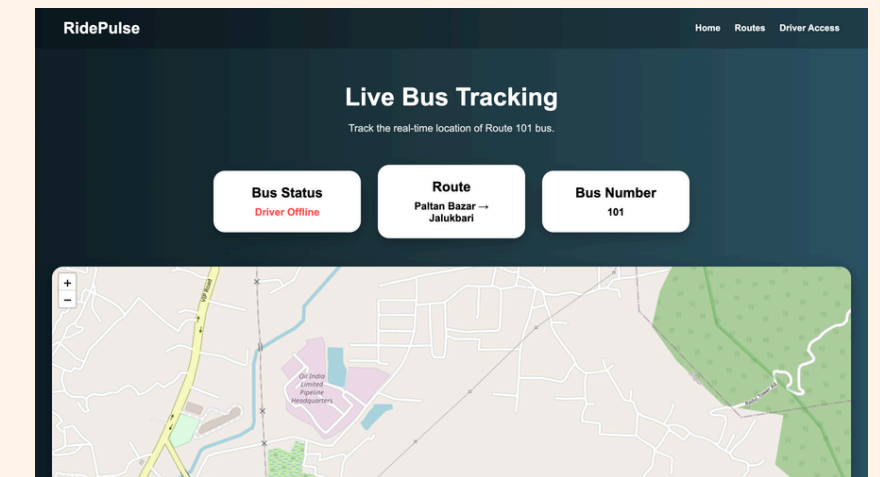
MODELS IMPLEMENTED



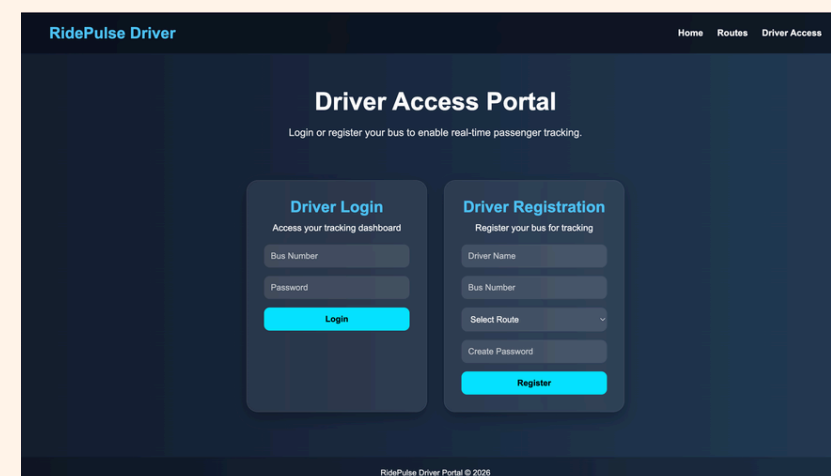
Home Page



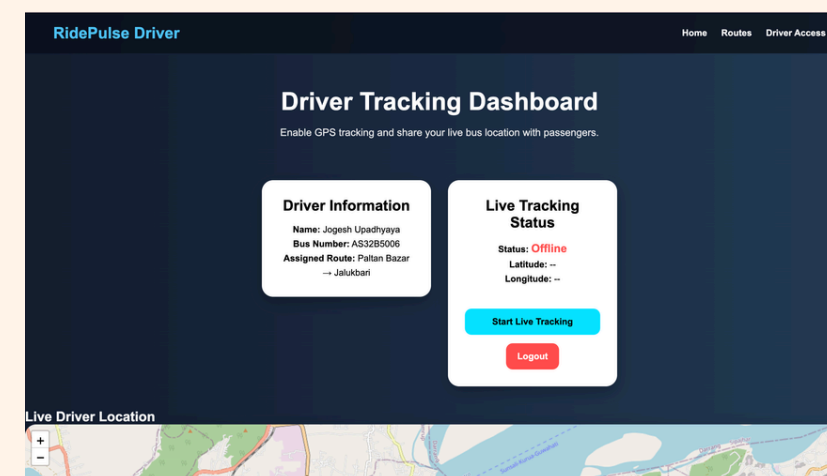
Routes Module



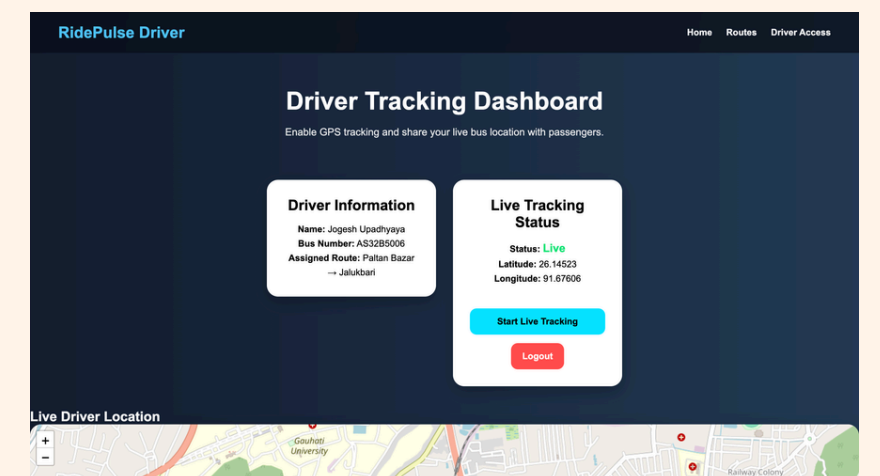
Passenger Tracking



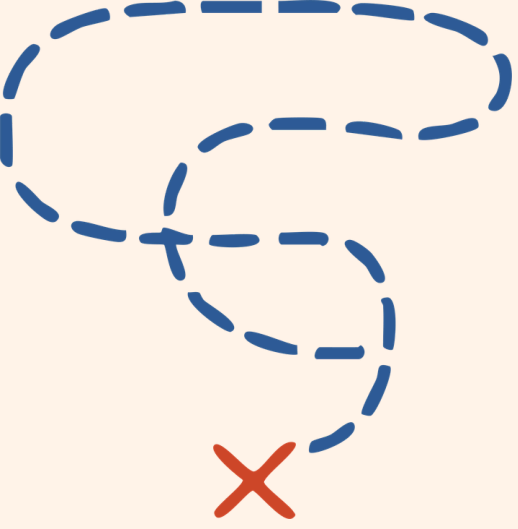
Driver Authorisation



Driver Dashboard
(offline)



Driver Dashboard
(online)



RESULTS AND ACHIEVEMENTS

The developed system was tested successfully on desktop and mobile devices.

Achievements

- Real-time GPS tracking achieved
 - Firebase synchronisation successful
 - Cross-device communication verified
 - Interactive map integration completed
 - Application deployed successfully
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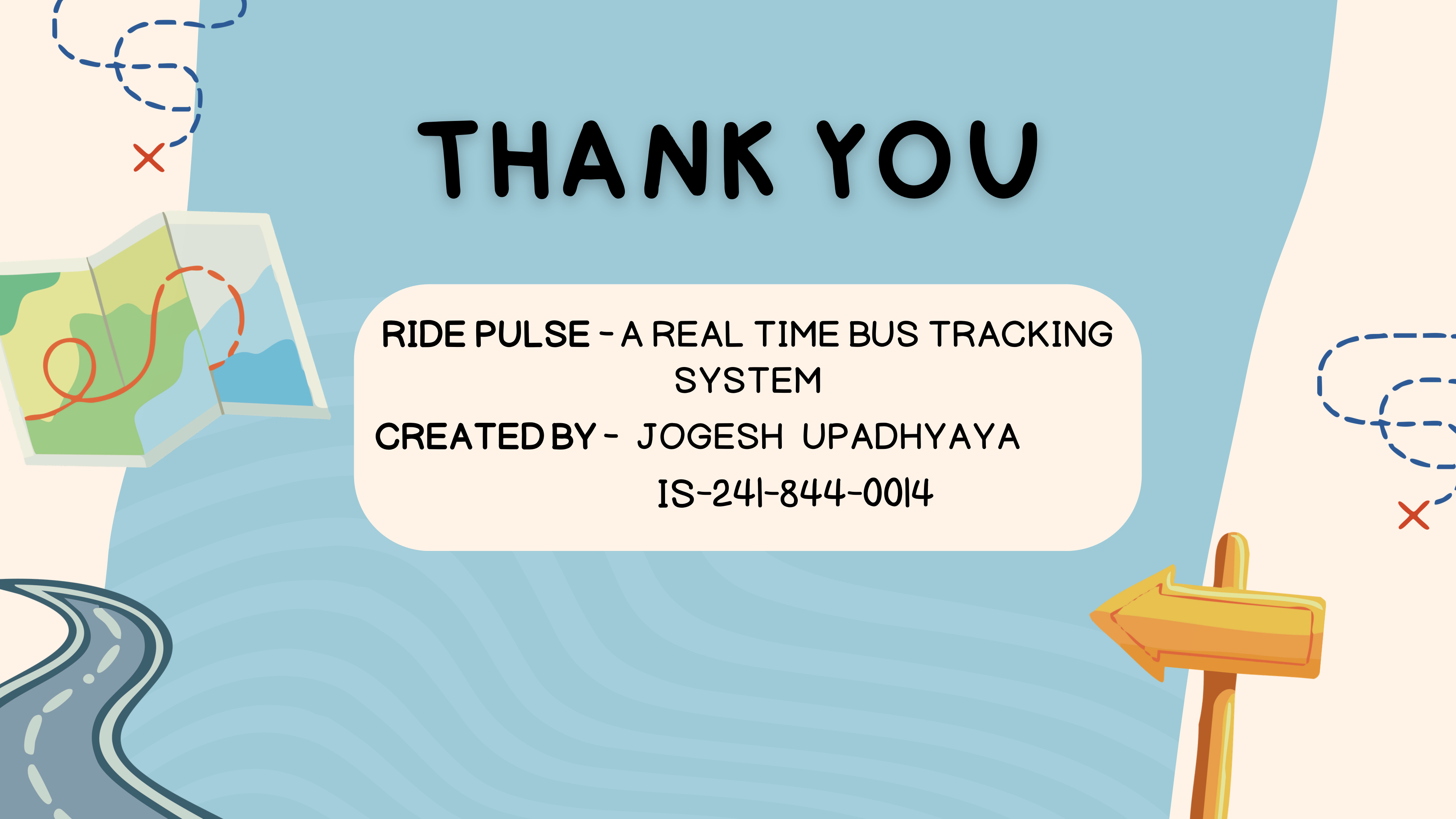
CONCLUSION AND FUTURE SCOPE

RidePulse successfully demonstrates a low-cost and accessible solution for real-time bus tracking using web technologies, GPS services, and cloud databases.

Future Scope

- Multi-bus support
- ETA prediction
- Mobile application
- Administrative dashboard





THANK YOU

RIDE PULSE - A REAL TIME BUS TRACKING
SYSTEM

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